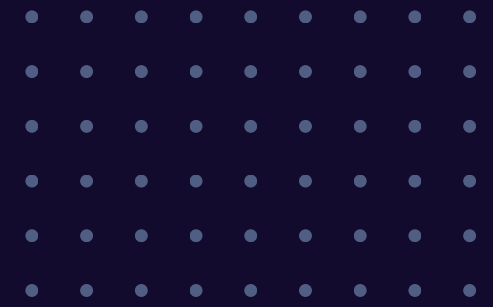




Climate Pattern

Clustering

Using K-Means & Hierarchical Clustering

Team Members:

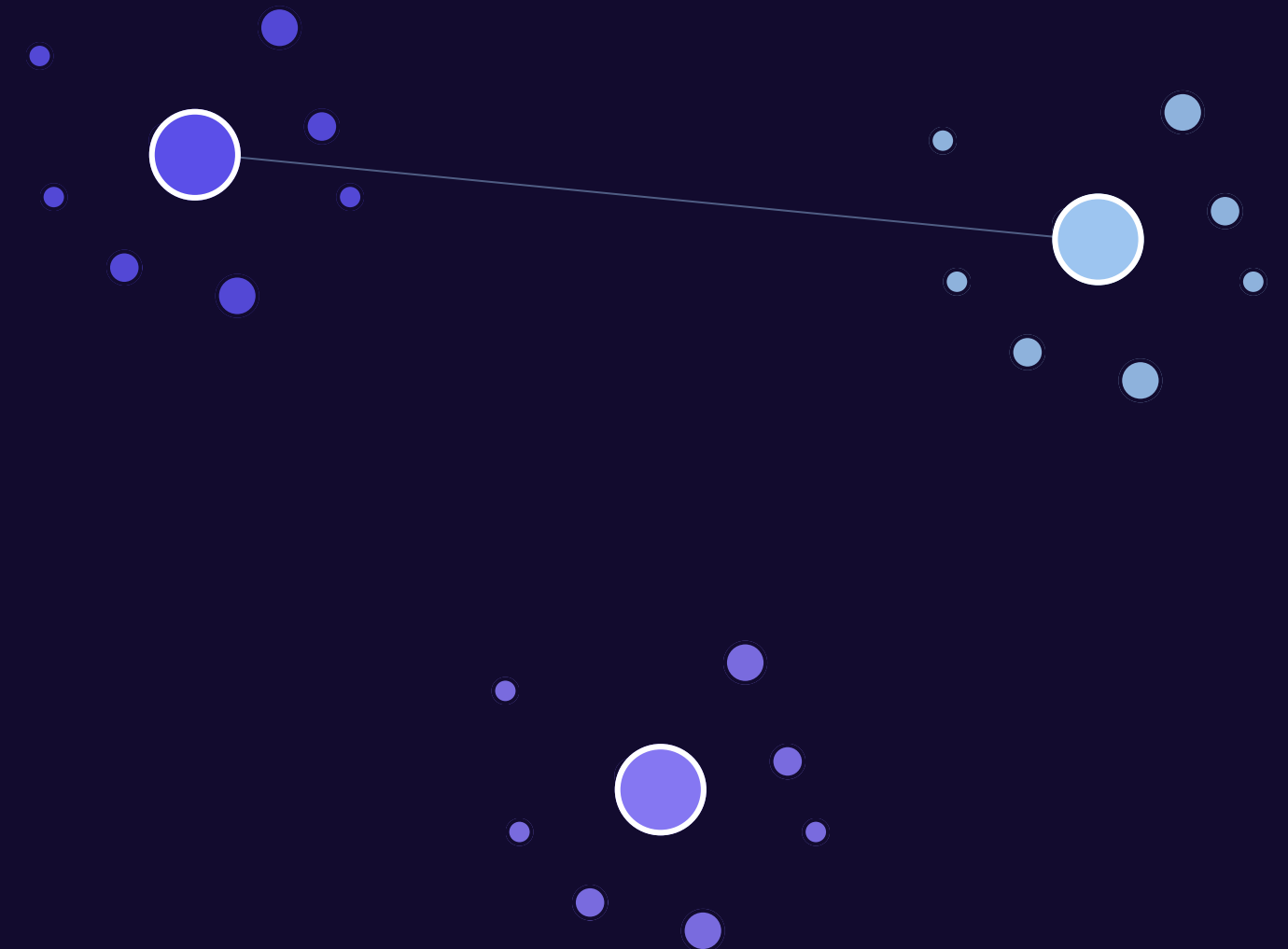
Praveen | 2520030406

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Machine Learning | Department of CSE

Guide: Dr. Shaik Asif

B.Tech Project Report | A.Y. 2026–27



Why Clustering for Climate Data?

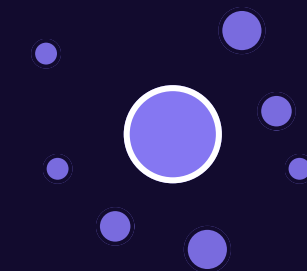
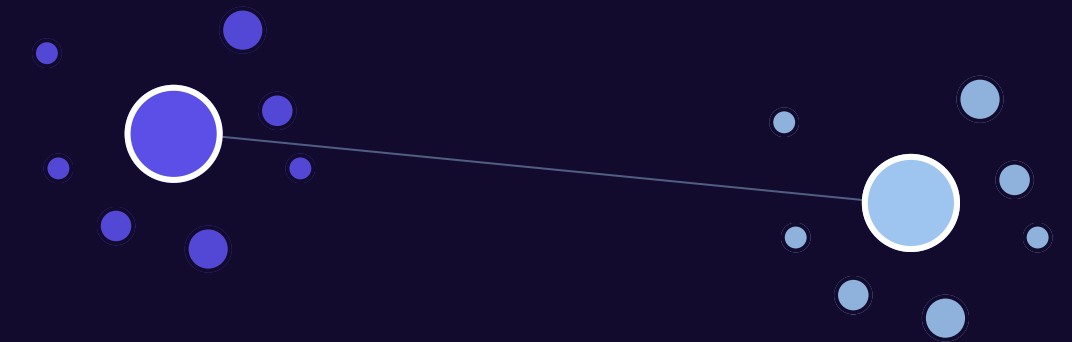
- Climate systems generate massive volumes of data — temperature, humidity, rainfall, wind speed, and pressure — across countless regions and time periods.
- This scale outpaces traditional, manual analysis methods.

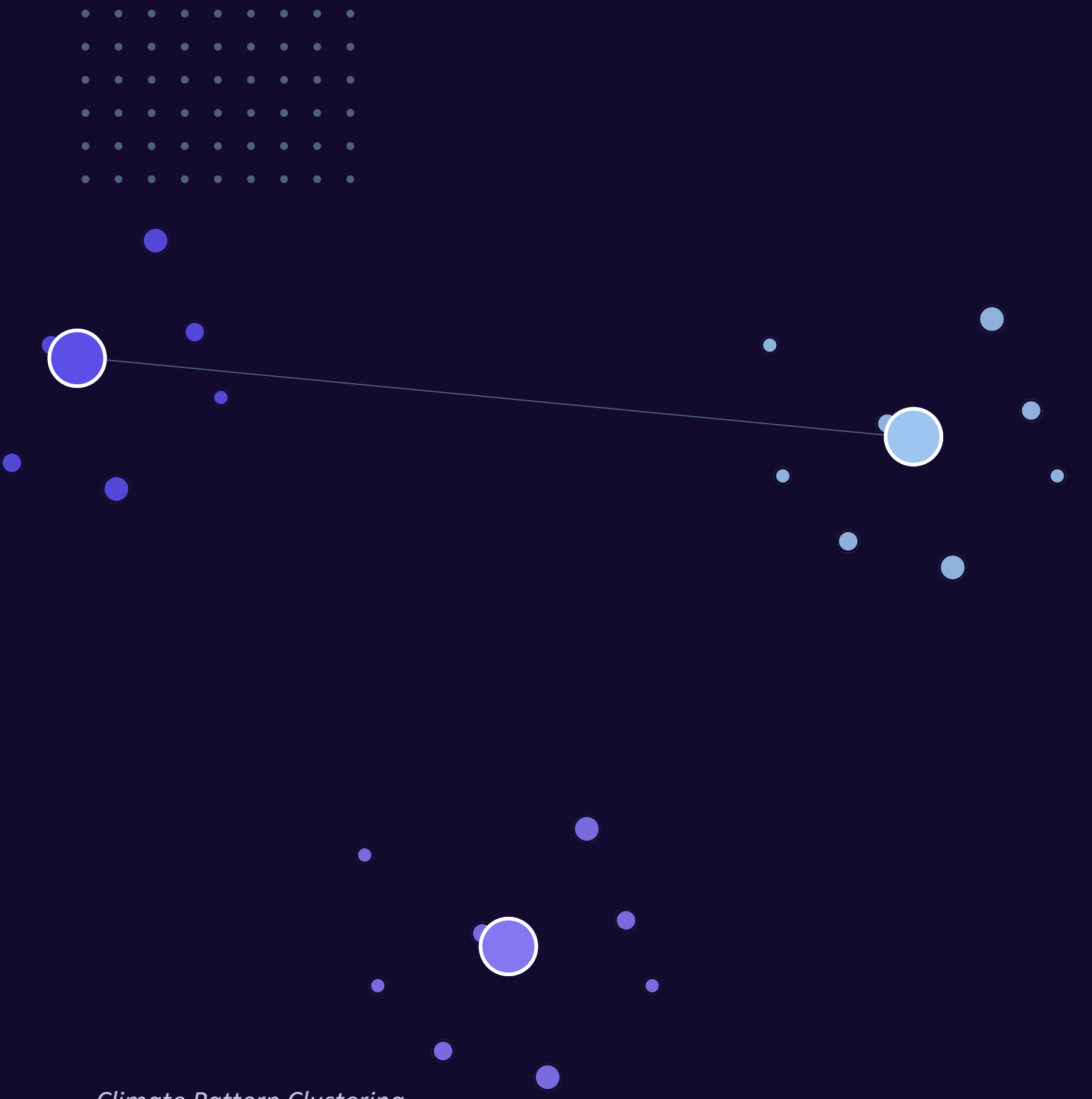
Key challenges this creates:

1. Difficulty spotting regional climate similarities
2. Slower environmental monitoring & forecasting
3. Limited support for agriculture & disaster planning

Traditional monitoring systems mainly report current conditions, rather than revealing which regions actually share similar climate behavior.

Our Goal: Group regions with similar climate characteristics using unsupervised machine learning.





Objectives

- Group regions by shared climate characteristics without predefined labels.
- Analyze temperature, humidity, rainfall, wind speed, and atmospheric pressure jointly.
- Implement K-Means and Hierarchical Clustering on the same climate dataset.
- Compare both methods and identify the more effective clustering approach.
- Support climate analysis, environmental monitoring, and disaster-management decisions.

Climate Pattern Clustering

Clustering Methods Overview



K-Means Clustering

Partitions climate records into K clusters by minimizing distance to each cluster's centroid. Fast and effective for well-separated climate zones.

Hierarchical Clustering

Builds nested clusters bottom-up and visualizes relationships via a dendrogram, revealing structure at multiple similarity levels.

Comparative Analysis

Both methods are evaluated using silhouette score, Davies–Bouldin index, and visual cluster separation to identify the better approach.

Both algorithms are applied to the same climate dataset to identify natural groupings among regions and compare which technique produces clearer, more actionable clusters.

Dataset Overview

Indian Climate Dataset (2024–2025)

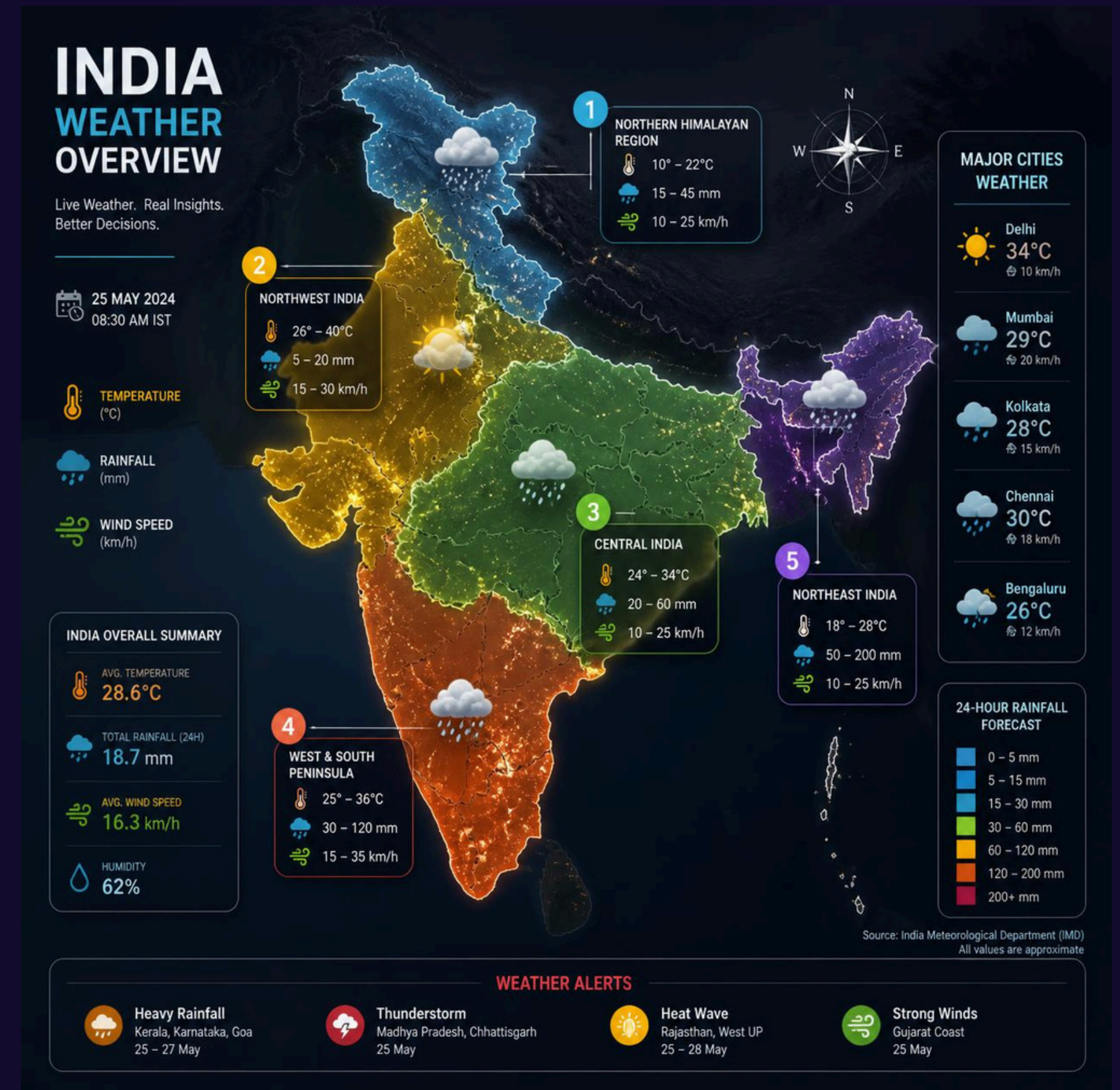
Source: Kaggle

Features:

1. Date
2. City
3. State
4. Maximum Temperature
5. Minimum Temperature
6. Average Temperature
7. Humidity
8. Rainfall
9. Wind Speed
10. Pressure
11. Cloud Cover
12. AQI
13. AQI Category

Dataset Size:

- 7,310+ records
- 13 columns



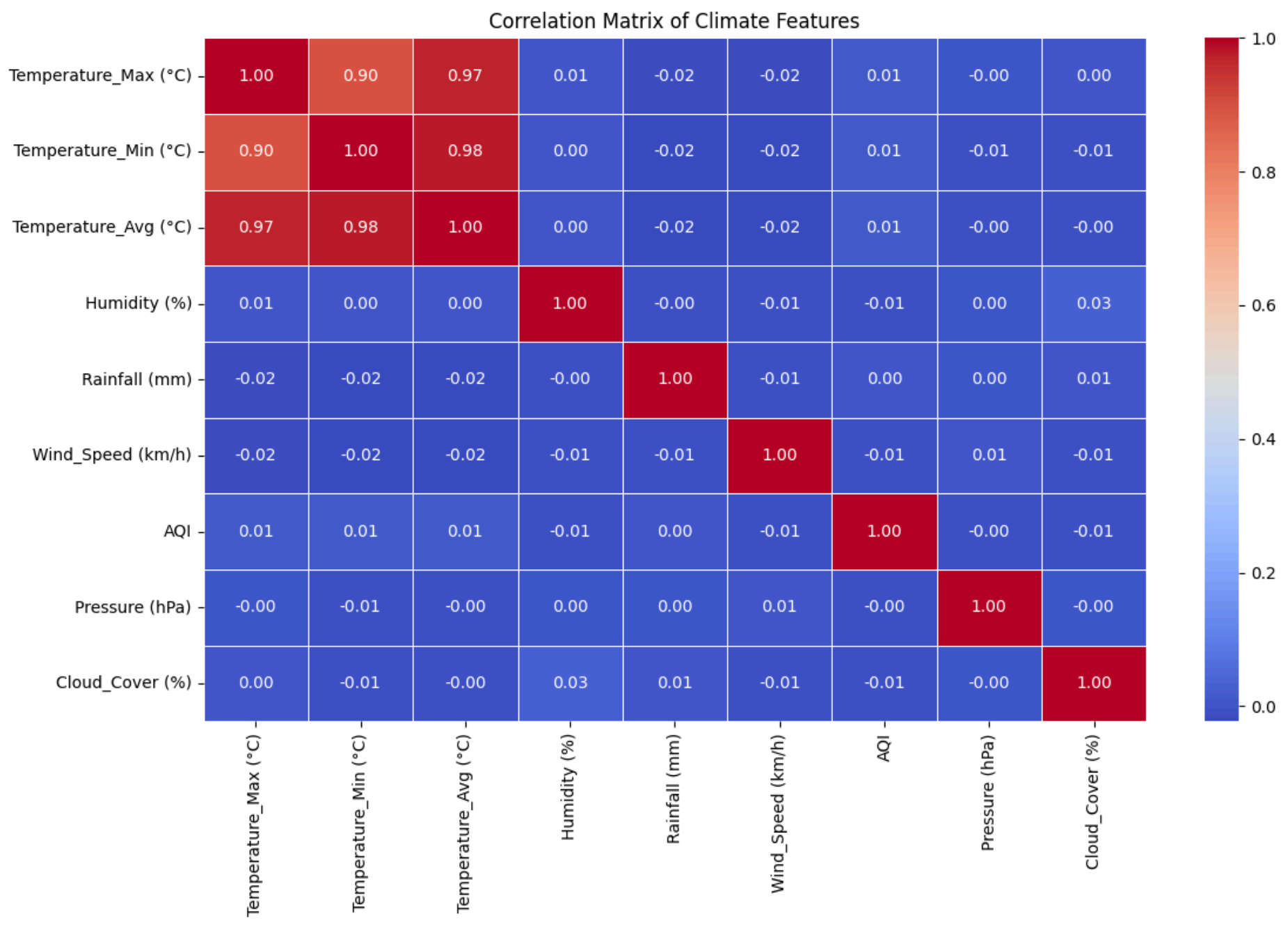
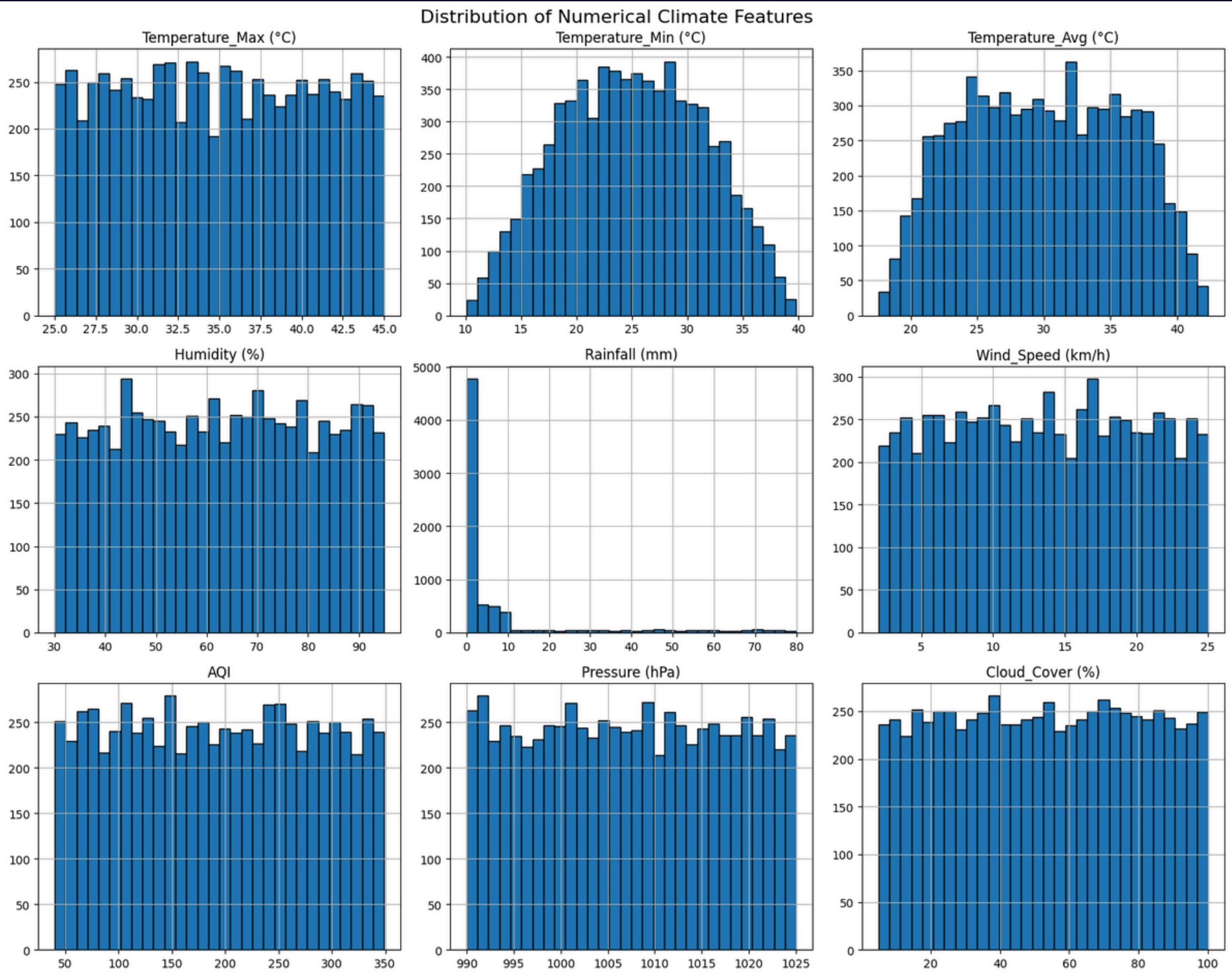
EDA Findings

Key Observations

- The dataset contains 7,310 climate observations across 13 attributes.
- Numerical features show different ranges and distributions, making feature standardization necessary before clustering.
- Temperature, humidity, rainfall, wind speed and pressure show different levels of variation across observations.
- Boxplots reveal potential extreme values, particularly in climate-related measurements.
- These extreme values should be evaluated carefully because they may represent genuine weather events rather than data errors.
- Correlation analysis helps identify relationships between climate variables and possible redundancy among highly related features.
- Since this is an unsupervised learning problem, there is no predefined target variable; clusters will be generated by the ML algorithms.

EDA confirms that the dataset contains sufficient variation in climate conditions to identify meaningful groups using clustering.

Graphs



Research Papers / Literature Survey

Research paper	Algorithm	Key Finding
Climate Classification India	K-Means	K-Means identified distinct climate types in India; the study found that k = 6 gave the highest silhouette score of 0.30 and improved climate classification.
Meteorological Conditions Spain	K-Means + Hierarchical Clustering	Clustering techniques effectively identified meaningful weather conditions in Spain; the study showed that comparing multiple algorithms with evaluation measures helps assess and select the quality of meteorological clusters.
Rainfall Clustering Mediterran	K-Means + Hierarchical Clustering	K-Means and Ward's hierarchical method produced similar classifications and successfully identified different rainfall distribution patterns over time. (ScienceDirect)

Development Environment & Tools

SOFTWARE

Programming Language	Python 3.x
Development Environment	Jupyter Notebook / Google Colab / VS Code
Operating System	Windows 10/11 or Ubuntu Linux
Key Libraries	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, SciPy
Dataset Format	Climate / Environmental dataset (CSV)

HARDWARE (MINIMUM)

**Intel Core
i3**

Processor

or higher

4 GB

Memory (RAM)

minimum required

500 MB

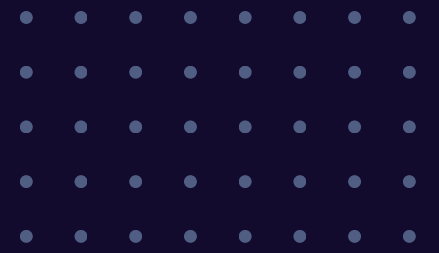
Free Storage

for datasets & environment

Lightweight Workflow

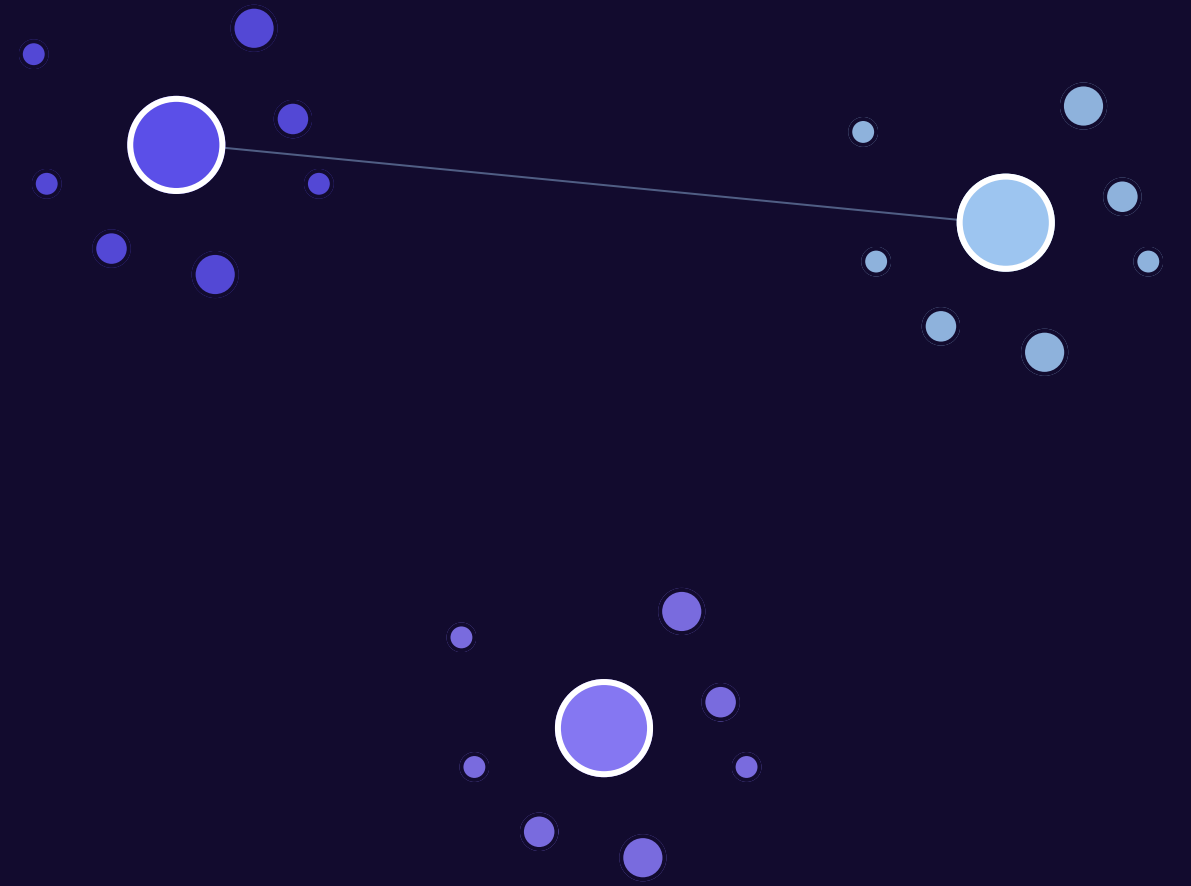
The clustering workflow (K-Means & Hierarchical Clustering, dendrogram visualization) runs comfortably on standard, entry-level hardware — no GPU required.

Conclusion



- Climate data holds valuable, hidden patterns that traditional monitoring alone cannot surface.
- This project groups regions by shared climate behavior using unsupervised machine learning — with no predefined labels.
- K-Means and Hierarchical Clustering are applied and compared on the same climate dataset.
- The results support climate analysis, environmental monitoring, agriculture planning, and disaster management.
- Together, the two methods show how unsupervised learning uncovers structure in large environmental datasets.

Turning raw climate data into meaningful, actionable patterns.



Thank You

Questions & Discussion

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